

Student Handout — Ohm's Law & Pack Math Quick Reference

Formulas

- $V = I \times R$
- $I = V / R$
- $R = V / I$
- $P = V \times I$ (power in watts)

Series vs Parallel (batteries)

Connection	Voltage	Capacity (Ah)
Series	Adds	Same as one unit (approx.)
Parallel	Same as one unit	Adds

Common pack examples

- **36V:** 6 × 6V in series (or other OEM arrangements — verify)
- **48V:** 6 × 8V in series (common) or 4 × 12V in series (common)

Meter reminders

Measuring	Probe jacks	Parallel or series with circuit?
Voltage	V / COM	Parallel
Resistance / continuity	Ω / COM	Power removed; across component
Current	A / COM (fused)	Series — advanced; prefer clamp meter

Practice problems (answers upside-down section for self-check)

1. 48V across $16\Omega \rightarrow I = ?$
2. 24A through $2\Omega \rightarrow V = ?$
3. Six 8V batteries in series $\rightarrow V = ?$

Answers

1. 3A
2. 48V
3. 48V